Engineering Structures 281 (2023) 115762

Contents lists available at ScienceDirect

Engineering Structures

journal homepage: www.elsevier.com/locate/engstruct

Role of moveable guide vane with various configurations in controlling the vortex-induced vibration of twin-box girder suspension bridges: An experimental investigation

Rui Zhou [a], Peng Lu [b], Xiaodong Gao [a], Yaojun Ge [b], Yongxin Yang [b,*], Lihai Zhang [c]

[a] College of Civil and Transportation Engineering, Shenzhen University, Shenzhen 518060, China
[b] State Key Lab for Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China
[c] Department of Infrastructure Engineering, University of Melbourne, VIC 3010, Australia

ARTICLE INFO

Keywords:
Twin-box girder
Moveable guide vane
Vortex-induced vibration
Control effectiveness

ABSTRACT

The purpose of this paper is to investigate the effects of a moveable guide vane with various configurations (*i.e.,* horizonal length and stiffness) in controlling the vortex-induced vibration (VIV) of a long-span suspension bridge with twin-box girder. First, a series of VIV tests of the 1:50 scaled ratio of sectional model with various guide vane configurations were performed. The vertical and torsional displacements of the twin-box girder as well as the vertical response of the windward and leeward guide vane measured to analyze the movement characteristics of the main girder and guide vanes, respectively. The test results show that, the moveable guide vane could significantly reduce the maximum vertical and torsional displacement responses of the main girder with an immovable guide vane or without a guide vane, especial under the most unfavorable wind attack angle of +3°. In addition, a moveable guide vane with relatively low stiffness could produce a better outcome of VIV control. Furthermore, the increase of the horizontal length of the guide vane could have a negative impact on the VIV control of the bridge. Importantly, the maximum vertical amplitude of the moveable guide vane is generally larger than that of the main girder, indicates that the leeward guide vane could play an important role in the VIV suppression when the stiffness of the guide vane is over certain threshold (*e.g.*, 19 g/mm).

1. Introduction

In recent years, the vortex-induced vibration (VIV) phenomenon of long-span suspension bridges were often observed all around the world, such as the well-known VIV event of Humen Bridge [1] and Yingwuzhou Yangtze River Bridge [2] in China, and Verrazano Bridge-Narrows Bridge in USA [3]. Although the VIV belongs to limited amplitude vibration, it often occurs at low wind speed, which could lead to the discomfort of drivers and long-term fatigue damage in suspension bridges [4,5]. When the VIV amplitudes of the main girder are larger than that of design requirement, passive aerodynamic countermeasures are commonly implemented for controlling the VIV due to their excellent economic and simple structure advantages [6,7]. Among different passive aerodynamic countermeasures of bridges, the guide vane is one of effective VIV countermeasure in the airflow separation around the box girder, such as the Stonecutters Island Bridge with guide vanes along the bottom plate [8] and the Gjemnessund bridge mounted with guide vanes [9]. However, the immoveable guide vane is sensitive to the VIV control of suspension bridges with various aerodynamic shapes, and its control effectiveness mainly depends on the configurations of the guide vane [10–13]. The moveable aerodynamic countermeasures are recently proposed to effectively improve the aerodynamic control effects of bridges [14,15,16], and the moveable guide vane combing the guide vane and spring in the design could has the potential to further improve the VIV control effect. However, the fundamental understanding of the influence of the moveable guide vane on VIV performance is still necessary before its wider application in suspension bridges with twin-box girder.

Through wind tunnel testing, theoretical modelling as well as field measurements, the control effect of the immoveable guide vane on the VIV of long-span bridges with various box girders have been extensively studied in the last decade. As for the single box girder, studies have shown that the guide vanes have has effect in suppressing the torsional VIV to some extent, while their abilities in reducing vertical VIV of the

* Corresponding author.
E-mail address: yang_y_x@tongji.edu.cn (Y. Yang).

https://doi.org/10.1016/j.engstruct.2023.115762
Received 6 July 2022; Received in revised form 7 January 2023; Accepted 2 February 2023
Available online 16 February 2023

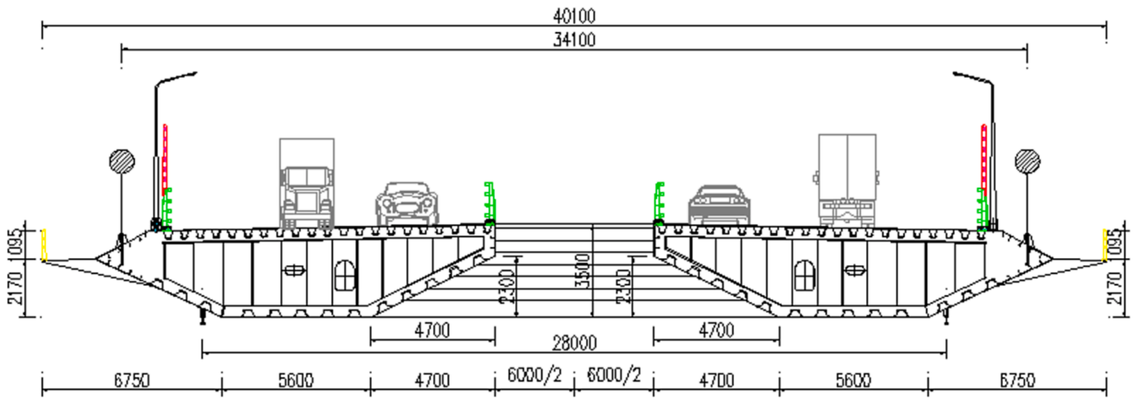

Fig. 1. The cross section dimension of the twin-box girder used in this study.

Table 1
Geometrical and mechanical parameters of the real bridge and sectional model.

Properties		Similarity ratio	Real bridge	Section model
Geometric scale	Length L/m	$\lambda_L = 1{:}50$	87	1.74
	Width B/m	$\lambda_L = 1{:}50$	40.1	0.802
	Height H/m	$\lambda_L = 1{:}50$	3.5	0.07
Equivalent mass	Mass/unit length m /kg/m	$\lambda_m = 1{:}50^2$	32,722.5	13.089
	Mass moment of inertia/unit length I_m/ kg·m^2/m	$\lambda_{jm} = 1{:}50^4$	5,689,970	0.91
Frequency	Vertical f_v /Hz	$\lambda_\omega = 25{:}1$	0.089503	2.238
	Torsional f_t/Hz	$\lambda_\omega = 25{:}1$	0.221171	5.529
Damping ratio	Vertical ξ_h/‰	$\lambda_\xi = 1{:}1$	5	5
	Torsional ξ_t/‰	$\lambda_\xi = 1{:}1$	5	4
Wind speed	vertical and torsional	$\lambda_w = 1{:}2$	0 ~ 40	0 ~ 20

Table 2
VIV testing cases used in this study.

Case	Number × dimension of spring	Total spring stiffness K (g/mm)	Vertical and horizontal length L (mm)	Wind attack angle (°)
1	–	–	–	0,±3
2	–	–	25 + 20	0,±3
3	3 × 0.6	19	**25 + 10**	0,±3
4	3 × 0.6	19	**25 + 20**	0,±3
5	3 × 0.6	**19**	**25 + 30**	0,±3
6	2 × 0.6 + 3 × 0.5	**25**	25 + 20	0,±3
7	2 × 0.7 + 2 × 0.5 + 1 × 0.6	**32**	25 + 20	0,±3

bridge is limited [17,18,19,20]. The VIV control effect of guide vane is closely correlated to the height, width and angle of guide vane itself, as well as the slot width for the twin-box girder [21,22,23,24,25]. The VIV tests and CFD simulation show that the guide vanes are detrimental to reduce the amplitudes of both heaving and torsional VIV of the 20 % slot width ratio [26,27]. Besides, the experimental study of Wang et al [28] show that the installation of guide vanes on wind fairings of the girder could intensify the vertical VIV responses of the triple-box girder under the wind attack angle of +3°. Since installing the immovable guide vanes may have adverse effects on VIV control of long-span bridges, the configuration of the moveable guide vanes play an important role in VIV control of long-span bridges, further research becomes necessary.

Through a series of wind tunnel tests, the purpose of the present study is to investigate the effectiveness of the moveable guide vanes on controlling the VIV of a super long-span suspension bridge with a twin-box girder. First, the VIV tests were conducted on the sectional models (geometrical scaled ratio of 1:50) of the twin-box girder with and without moveable guide vanes. Two critical parameters of the moveable guide vanes, *i.e.*, representative horizontal length (*e.g.*, 10 mm, 20 mm and 30 mm) and stiffness (*e.g.*, 19.5, 22.3 and 42.7 g/mm), were then investigated. Finally, the movement characteristics of the windward and leeward guide vane are also discussed. In summary, the present study could contribute to the optimization design of the moveable guide vane on VIV performance of bridges instead of the immoveable guide vane.

2. Methods

2.1. Experimental set-up of VIV tests

A super long-span suspension bridge with a span arrangement of 580 + 1756 + 630 m was selected in this study, in which the dimension of the twin-box girder is 40.1 m wide and 3.5 m height with the central gap of 0.6 m. The modal analysis results show that the first vertical and torsional frequency of the actual bridge is 0.0895 Hz and 0.2211 Hz, respectively. The cross-section dimension of the twin-box girder is shown in Fig. 1, and the structural parameters of the actual bridge and sectional model with a scale ratio of 1:50 are listed in Table 1. The

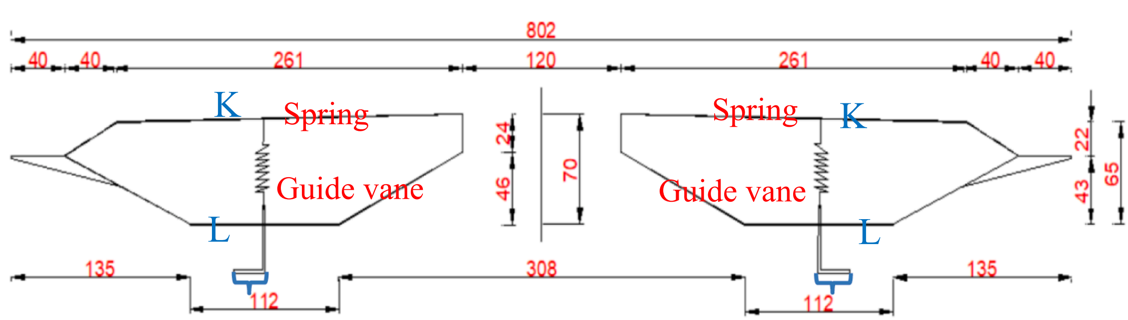

Fig. 2. The design of moveable guide vane (unit: mm).

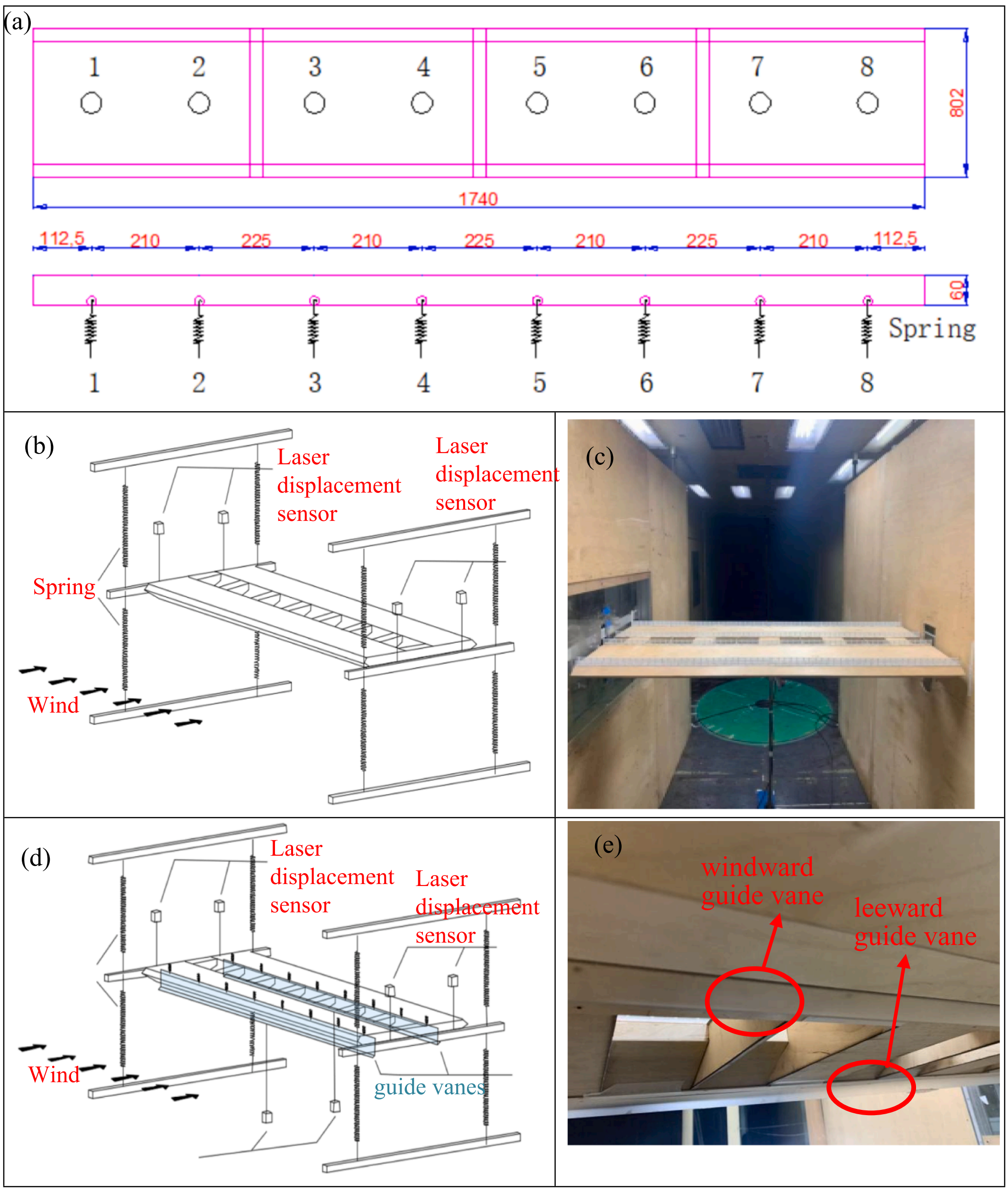

Fig. 3. The sectional models used in VIV tests. (a) the arrange of the moveable guide vanes; (b-c) the immoveable guide vanes; (d-e) the moveable guide vanes.

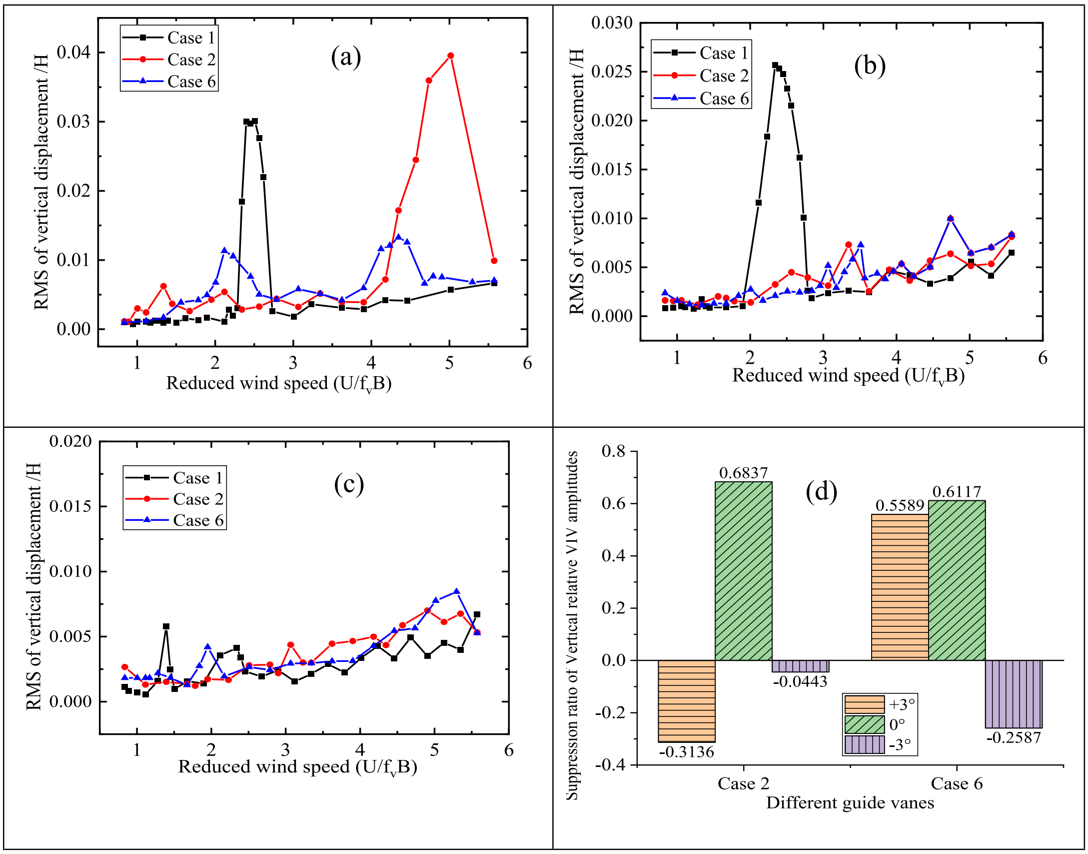

Fig. 4. Vertical VIV displacement responses of the bridge with various guide vanes:(a) α=+3°,(b) α=+0°,(c) α = −3°, and (d) suppression ratios of three wind attack angles.

geometric dimensions, mass characteristics, fundamental frequencies, and structural damping of the sectional model were estimated based on the 1:50 geometrical scale ratio.

2.2. VIV tests of a twin-box girder without and with the immoveable guide vane

As shown in Fig. 2, the moveable guide vane is composite of the immoveable guide vane and the spring, in which many spring are connected by the top plate of the main girder and the upper end of the guide vanes. Based on the Den Hartog optimization theory of the tuned mass damper (TMD) parameters [29,30,31], the stiffness and mass of the moveable guide vane are determined by the optimal parameters formulas:

$$m_2 = \mu m_1,\ \gamma = \frac{1}{1+\mu},\ \omega_2 = \frac{\mu}{1+\mu} \times \omega_1,\ k_2 = {\omega_2}^2 \times m_2 \tag{1}$$

where m_1 and ω_1 are the mass and frequency of the target section model, respectively. γ is the most optimal frequency ratio. m_2, k_2 and ω_2 are the mass, stiffness and frequency of the moveable guide vane, respectively. In addition, the length of the guide plate is determined by the mass information of the moveable guide vanes used in the present study. In order to comprehensively investigate the influence of the spring stiffness (K) and horizontal length (equivalent to mass, L) on the VIV performance of the twin-box girder bridge, three representative horizontal lengths (L = 10 mm, 20 mm and 30 mm) and three representative spring stiffnesses (K = 19.5 g/mm, 22.3 g/mm and 42.7 g/mm) were designed for the sectional model. As listed in Table 2, there are 7 testing cases with the combination of different horizontal lengths (Case 3 to 5), and spring stiffnesses (Case 5 to 7). There is no guide vane installed in Case 1 with three attack angles of +3°, 0° and −3°, while Case 2 is the testing case with the immoveable guide vane. As shown in Fig. 3(a), there are 8 springs along the length of section model with the same interval of 0.2 m, and the total length of the section model is 1.74 m.

A series of VIV tests were conducted in the TJ-2 Boundary Layer Wind Tunnel at Tongji University, as described in the experimental setup of Fig. 3(a-b). The geometric scale of the test section is 15 m length × 3 m length × 2.5 m height, the wind speed ranges from 0.5 to 68 m/s, and the turbulent flow is smaller than 0.46 %. In Cases 1 & 2, there are 4 laser displacement transducers were used for measuring the vertical and torsional displacement responses of the main girder due to vortex-shedding excitation, respectively. As for Cases 3–7 with the moveable guide vanes, two laser displacement transducers are also used to measure the vertical displacement responses of the windward and leeward guide vane in Fig. 3(c-d), respectively.

3. Results

3.1. Comparison of VIV responses without and with the guide vanes

Vertical displacement responses of the bridge without and with the immoveable and moveable guide vanes under three wind attack angles (α = -3°, 0° and +3°) are described in Fig. 4 (a-c), respectively. As shown in Fig. 4, the RMS values of vertical VIV displacement of Case 1 remains relatively small (<0.0065) when U/f$_v$B increases from 3 to 6, and the vibration is forced vibration caused by characteristic turbulence, not belong to the VIV. Under the wind attack angle of +3°, a movable guide vane (Case 6) can significantly reduce the maximum normalized root mean square (RMS) of vertical displacement by threefold for case without installation of the guide vane (Case 1) and by fourfold for an

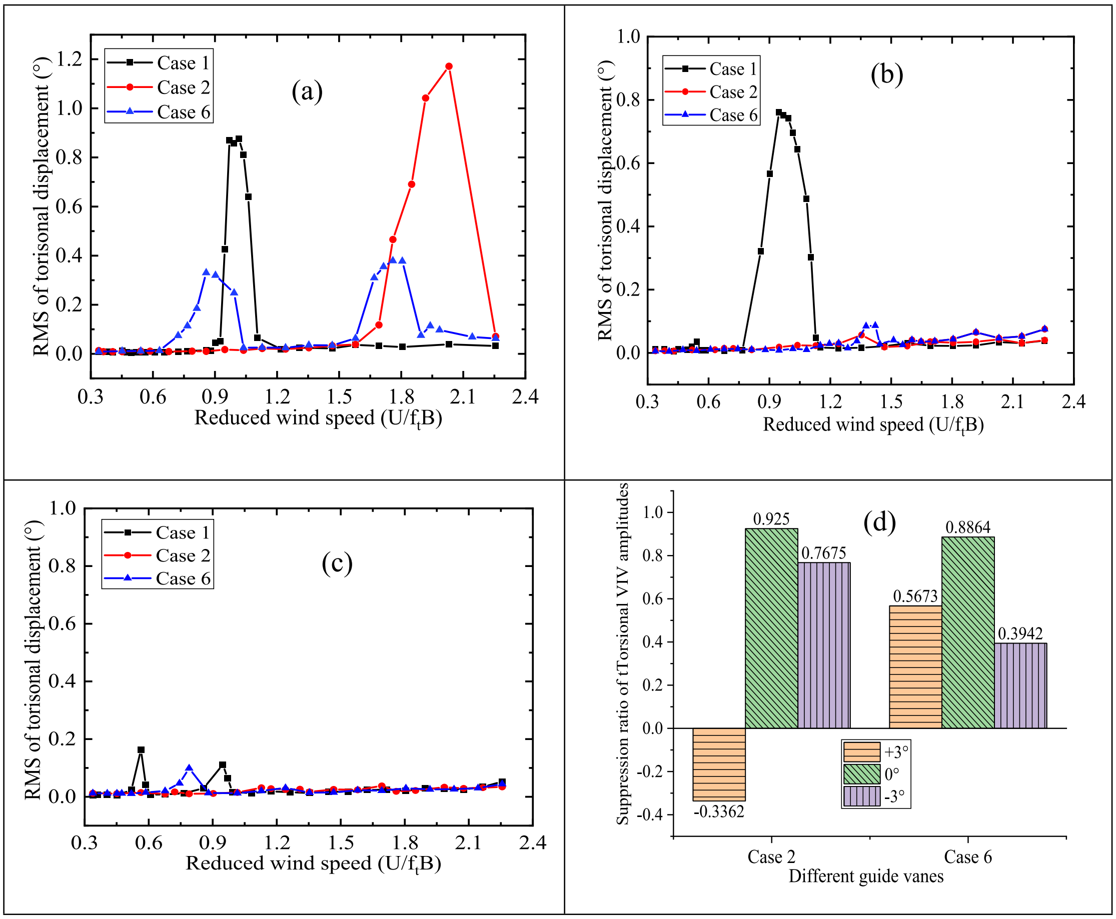

Fig. 5. Torsional VIV displacement responses of the bridge with various guide vanes: (a) α=+3°, (b) α=+0°,(c) α = −3°, and (d) suppression ratios of three wind attack angles.

immovable guide vane (Case 2). Importantly, it suggests that the reduced wind speed U/f_vB of the Case 2 and Case 5 of the peak amplitude are about 2.5 and 5, which are larger than those of the Case 1 with $U/f_vB = 2$. Both the immovable and moveable guide vane have significant impact on the RMS reduction of vertical displacement under the wind attack angle of 0°, and there is no obvious difference can be seen when the α = -3°. Fig. 4(d) compares the suppression ratio (φ) of vertical VIV amplitude under immoveable (Case 2) and moveable guide vane (Case 6), respectively. $\varphi = \frac{A_1 - A_o}{A_1} \times 100\%$, where the A_1 and A_o is the maximum values of VIV displacement of the Case 1 (no guide vane) and other cases, respectively. It shows that the values of φ for Case 2 and Case 6 are −31 % and 56 %, respectively. In summary, the twin-box girder with the moveable guide vane produces the best vertical VIV performance in comparison to the immoveable one, especial under the most unfavourable wind attack angle of +3°.

Fig. 5 compares the torsional displacements' RMS of without and with the guide vanes under three wind attack angles of −3°, 0° and +3°, respectively. Similar to that of vertical VIV responses, it shows that, under the wind attack angle of +3°, a movable guide vane (Case 6) can significantly reduce the maximum torsional displacement by more than twofold for the case without the guide vane (Case 1), and by more than threefold for an immovable guide vane (Case 2). For the wind attack angle of 0°, the immoveable guide vane could have the significant negative impact on the torsional displacements of the bridge, while the torsional displacement under wind attack angle of −3° are limited. In addition, although the suppression ratios of Case 2 are slightly larger than those of the Case 6 under the α = 0° and −3°, the suppression ratios of Case 2 and Case 6 are around −34 % and 57 %, respectively, under the α= +3°in Fig, 5(d). The results indicate that the installation of moveable guide vane can effectively decrease the maximum torsional VIV responses, whereas the immoveable guide vane also has negative effect on torsional VIV responses of the bridge.

3.2. VIV responses under different horizontal lengths of a moveable guide vane

In order to compare the VIV control effect of the moveable guide vanes with various horizontal lengths, vertical and torsional VIV responses of the bridge with three different horizontal lengths under three wind attack angles are described in Fig. 6 and Fig. 7, respectively. Fig. 6 (a) shows that, under the wind attack angle of α= +3°, the RMS of vertical displacement gradually decreases with the increase of the horizontal length of a guide vane. Compared with the 0° and −3° wind attack angle, the aerodynamic shape of the twin-box girder approaches more bluff body under the positive +3° wind attack angle. The different horizontal lengths of the moveable guide vane could lead to the different separation locations of airflows around the guide vane, and thereby result in different VIV control effects. The results in Fig. 6(b-c) show that the horizontal length of the guide vane has the limited influence in the RMS of vertical displacement responses of the bridge under the α = 0° and −3°. In addition, Fig. 6(d) shows that the suppression ratio of vertical displacement response of the relatively short horizontal length of a guide vane (*e.g.,* Case 3) is higher than that of Case 4 and Case 5. Both the suppression ratios of the torsional VIV under the α=+3° and α=+0° are slight larger than those of the vertical VIV, while the suppression ratios of the torsional VIV under the α = -3° are opposite to

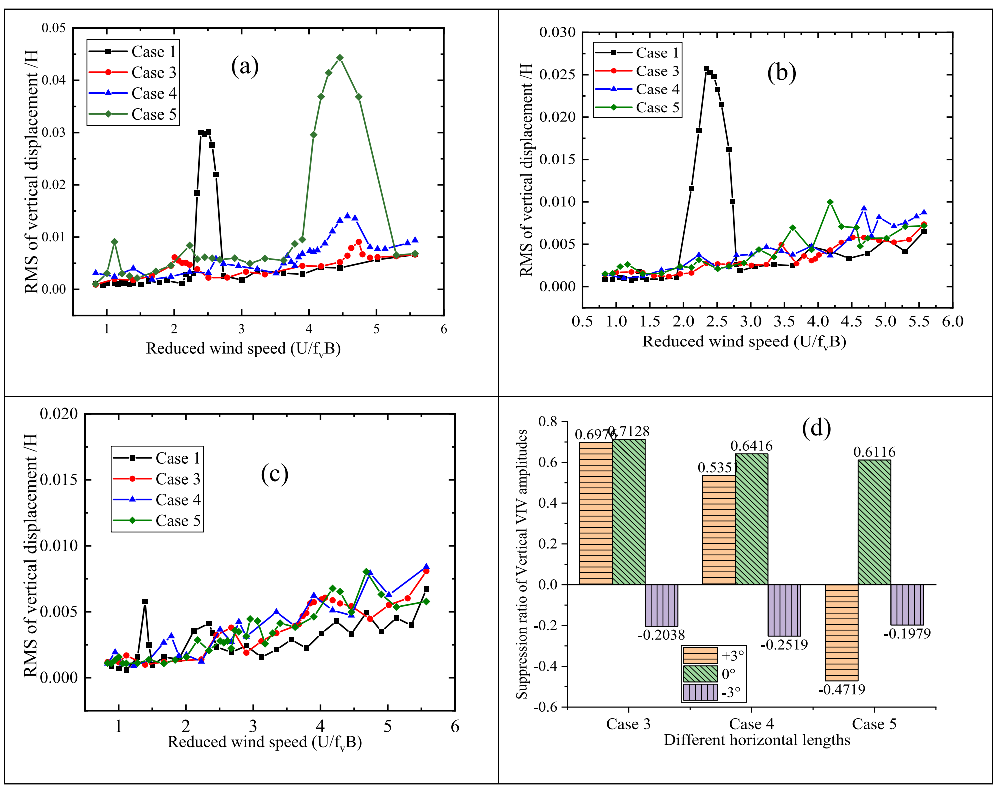

Fig. 6. Vertical VIV responses of the bridge with various horizontal lengths of a guide vane: (a) α=+3°,(b) α=+0°,(c) α = -3°, and (d) suppression ratios of three wind attack angles.

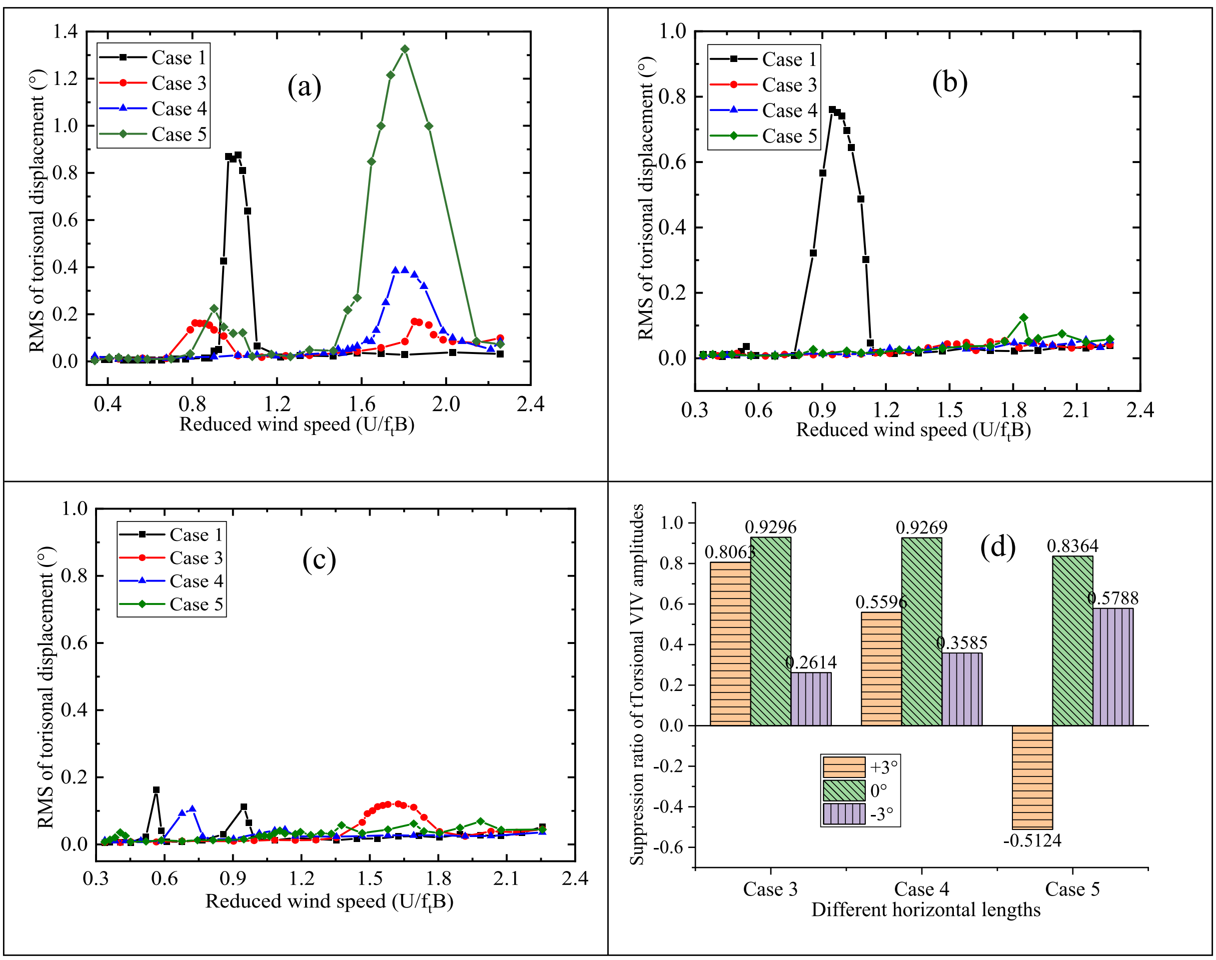

Fig. 7. Torsional VIV responses of the bridge with various horizontal lengths of a guide vane: (a) α=+3°,(b) α=+0°,(c) α = −3°, and (d) suppression ratios of three wind attack angles.

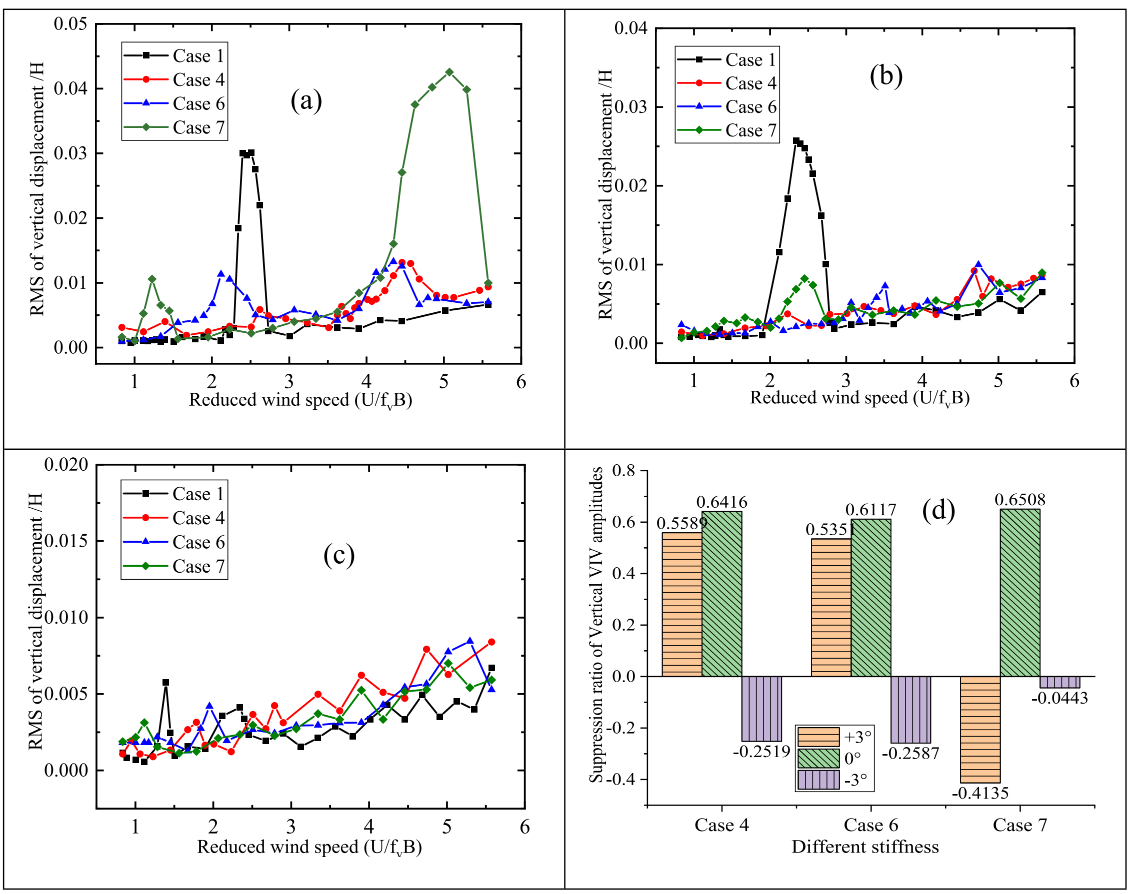

Fig. 8. Vertical VIV responses of the bridge with various spring stiffnesses: (a) α=+3°, (b) α=+0°, (c) α = −3°, and (d) suppression ratios of three wind attack angles.

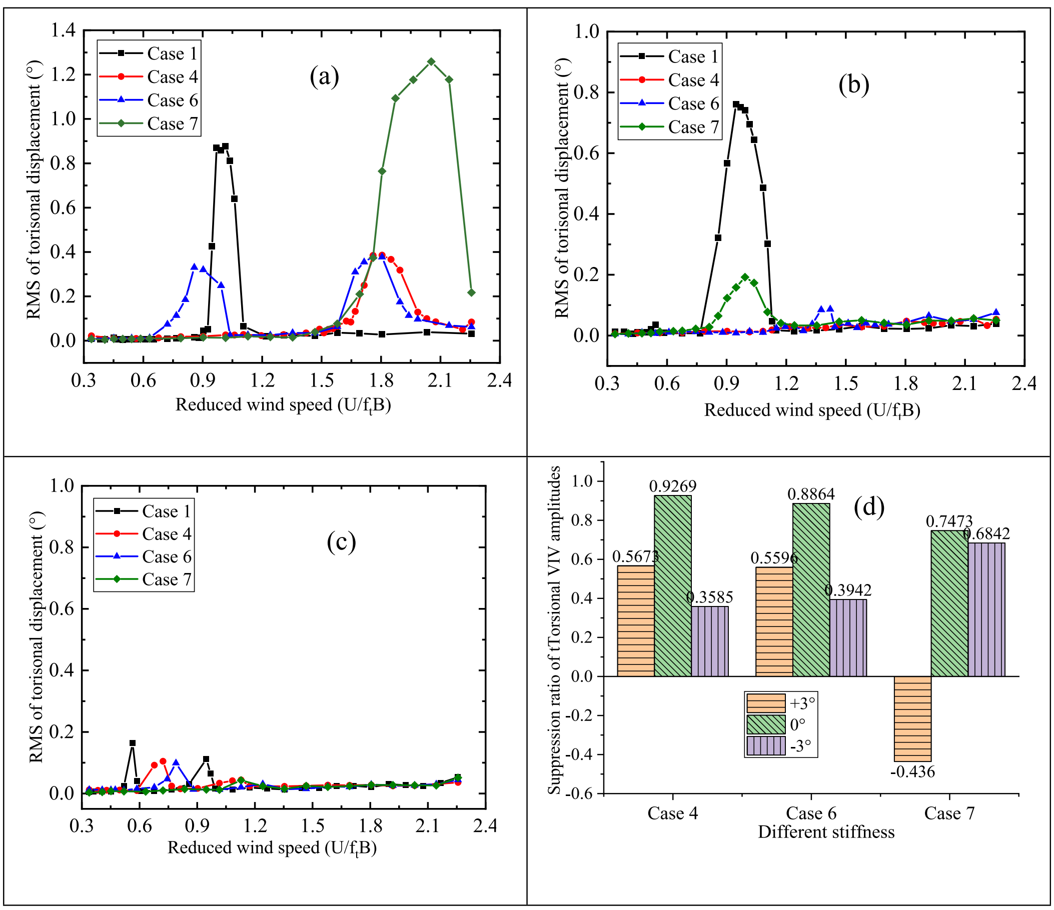

Fig. 9. Torsional VIV responses of the bridge with various spring stiffnesses: (a) α=+3°, (b) α=+0°,(c) α = −3°, and (d) suppression ratios of three wind attack angles.

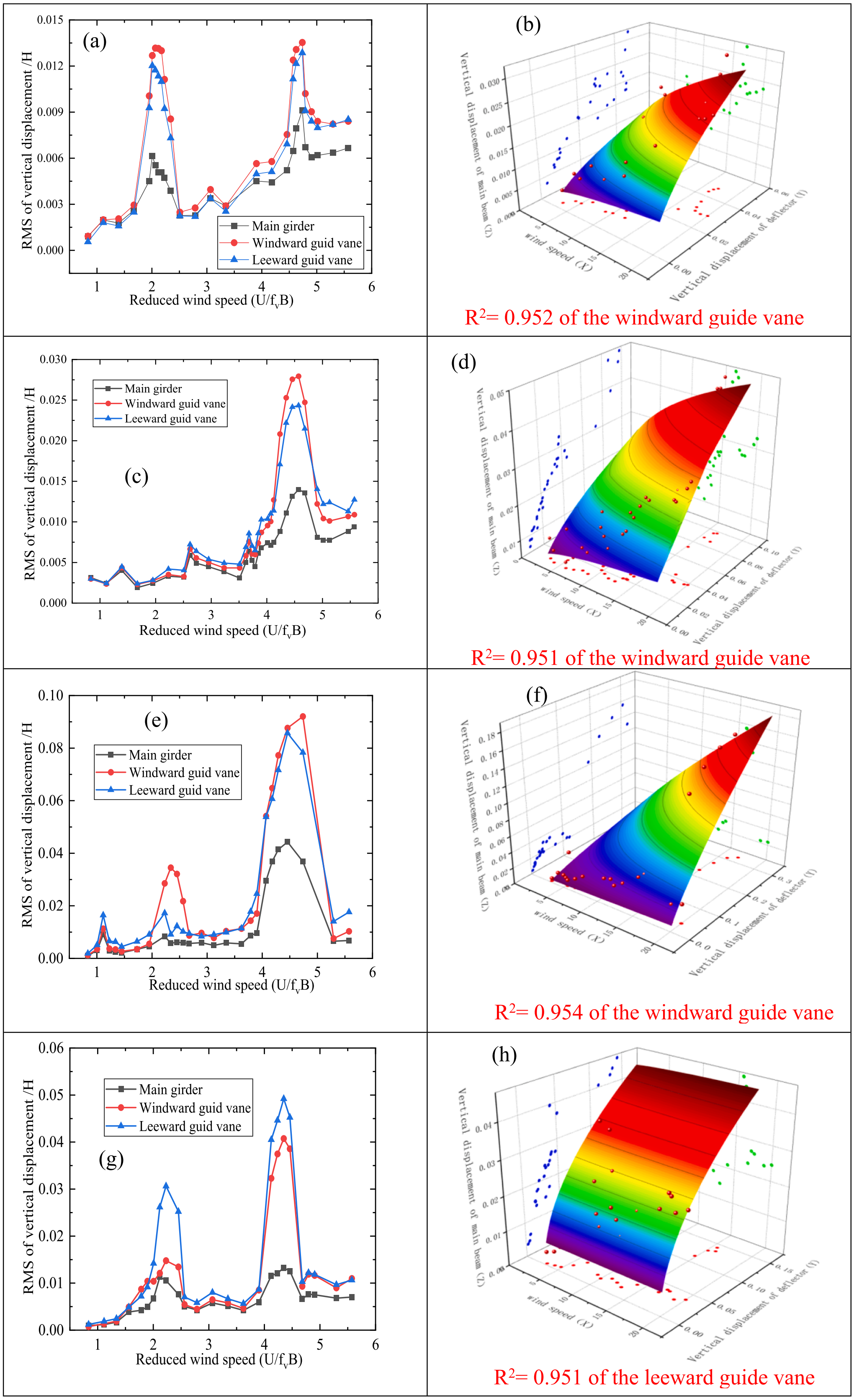

Fig. 10. Vertical displacement and movement characteristics of the moveable guide vanes under α=+3°: (a-b) Case 3; (c-d) Case 4; (e-f) Case 5; (g-h) Case 6; (i-j) Case 7.

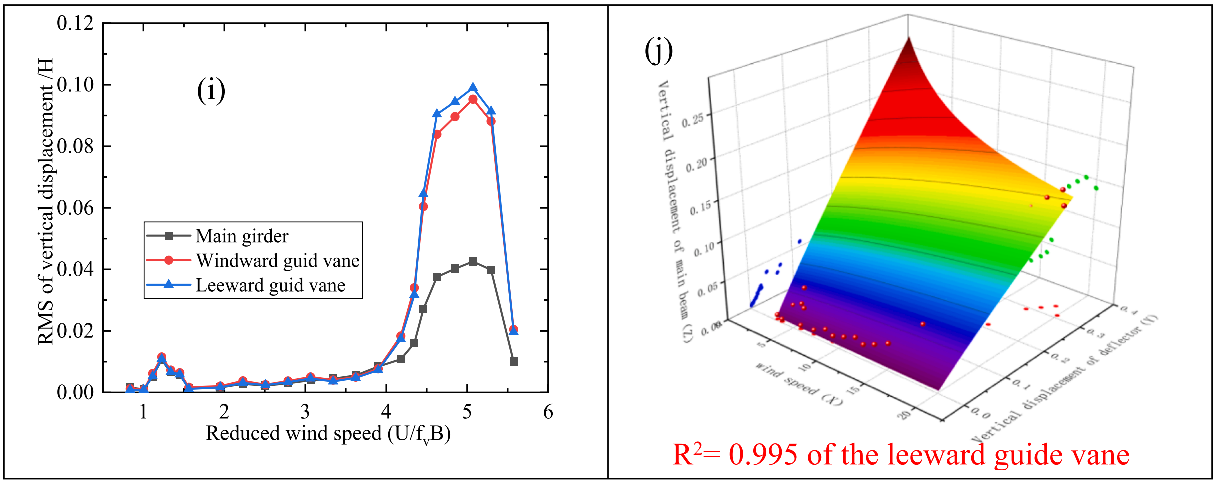

Fig. 10. (*continued*).

those of the vertical VIV. Therefore, the effects of various horizontal lengths of a guide vane on the torsional displacement responses of the bridge are similar to that of vertical displacement responses under the α=+3° and α=+0° as demonstrated in Fig. 7.

3.3. VIV responses under various stiffnesses of a moveable guide vane

The influence of the moveable guide vane with various spring stiffnesses in the RMS of vertical and torsional VIV displacement responses of the bridge is shown in Fig. 8 and Fig. 9, respectively. The moveable guide vane with low spring stiffness (*e.g.,* Case 4) could produce better outcomes in reducing the vertical and torsional VIV displacement responses compared to that with high spring stiffness (*e.g.,* Case 6 and Case 7).

3.4. Movement characteristics of the moveable guide vane

The mapping relationship among the wind speed, the vertical displacement of the main girder and the moveable guide vane were fitted by the Levenberg-Marquardt and Universal Global Optimization (LM-UGO) method using the fitted model of Z = p1*X^p2*Y^p3, where *X* is the wind speed, *Y* is the vertical displacement of the moveable guide vane, *Z* is the vertical displacement of the main girder. The purpose of global optimization is to obtain the best (global) solution through multiple local and global optimization. LM - UGO method has proven capabilities of obtaining the global optimal solution with fast speed and high accuracy so that it is implemented to establish the mapping relationship without the need for appropriate initial condition [32]. As presented in the Fig. 10, we can find the fitted model can perfectly satisfy the demands since all these R-Squares of these different cases are larger than 0.95. Fig. 10 also shows the vertical displacement responses of the moveable windward and leeward guide vane (from Case 3 to Case 7) under the most unfavourable α=+3°. For all the cases, the vertical displacement responses of the windward and leeward guide vanes are much larger than those of the main girder, especial in the lock-in regions. With the increase of the horizontal length of the guide vane, the vertical displacement response of the guide vanes could significantly increase, and the peak vertical relative displacement of the windward guide vane is larger than those of the leeward vane. In addition, with the increase of the stiffness, the vertical displacements of the guide vanes could increase significantly, and the peak vertical displacements of the leeward guide vane are larger than those of the windward guide vane. Furthermore, the vertical amplitudes of the moveable guide vanes are larger than that of the main girder, which is helpful to reduce the vibration of the bridge to some extent. The leeward guide vane could play a more important role in the VIV suppression when the total stiffness of the moveable guide vane is larger than 19 g/mm. In the lock-in wind speed region of VIV, the horizontal displacement of the moveable guide vanes under the action of wind could be neglected since the small notch could restrict the horizontal motion of the moveable guide vanes. The movement of the moveable guide vanes can result in energy dissipation of the section model system in the lock-in wind speed of VIV. The effect of the movement acceleration of guide vanes should be studied in future VIV tests through installing small acceleration sensors on the movable guide vanes and section model, respectively.

In the design of the moveable guide vane, we aim to suppress the first vertical VIV to calculate the optimal mass and stiffness of the moveable guide vane. The mass ratio μ and frequency ratio γ between the moveable guide vane and the cross-sectional model (in both vertical and torsional direction) of different testing cases are calculated and the results are shown in Table 3, respectively. The mass ratios gradually increase with the increase of the length or the stiffness of the movable guide vane due to the mass increment of the guide vane and spring, respectively. Meanwhile, the vertical frequency ratios gradually decrease with the increase of the length or the stiffness of the movable guide vane due to the vertical frequency reduction of the movable guide vanes and the number increment of spring in the VIV tests, respectively. The control mechanisms of the Case 4 to Case 7 are mainly the vertical VIV suppression effects of both the vertical TMD and guide vanes, while

Table 3
Control mechanism of different testing cases investigated in this study.

Case	Direction	μ	Theoretical γ	Measured γ	Control mechanism
Case 3	Vertical	0.0181	0.982	0.981	TMD
Case 3	Torsional	0.0181	0.982	0.393	Guide vane
Case 4	Vertical	0.0225	0.978	0.928	TMD + Guide vane
Case 4	Torsional	0.0225	0.978	0.376	Guide vane
Case 5	Vertical	0.0269	0.974	0.775	Guide vane + TMD
Case 5	Torsional	0.0269	0.974	0.314	Guide vane
Case 6	Vertical	0.0239	0.977	0.903	TMD + Guide vane
Case 6	Torsional	0.0239	0.977	0.358	Guide vane
Case 7	Vertical	0.0252	0.975	0.798	Guide vane + TMD
Case 7	Torsional	0.0252	0.975	0.322	Guide vane

the Case 3 can exert the TMD effect. However, the torsional frequency ratios are much smaller than those of the vertical frequency ratios, which indicates that the movable guide vanes only have the similar effect of torsional VIV suppression of immovable guide vanes. It should be mentioned that this paper mainly focuses on the experimental study on the roles of a moveable guide vane with various configurations in controlling the VIV of twin-box girder suspension bridges, while the aerodynamic mechanism will be included in the future studies.

4. Conclusion

In present study, a series of VIV tests are conducted to study the effectiveness of the moveable guide vanes in controlling the VIV performance of a super long-span suspension twin-box girder bridge. The following are some major findings:

- The moveable guide vane could significantly reduce the maximum vertical and torsional displacement responses of the cases without and with immoveable guide vane by threefold and fourfold, respectively, under the wind attack angle of $+3^\circ$. However, the influence of the moveable guide vane is limited under the wind attack angles of 0° and -3°.
- A more flexible moveable guide vane (*i.e.,* relatively low spring stiffness) could improve the VIV performance of the bridge by reducing its maximum vertical and torsional displacement responses.
- Under the wind attack angle of $\alpha = +3^\circ$, the maximum vertical and torsional displacement responses gradually decrease with the increase of the horizontal length of the guide vane. However, the influence of the horizontal length of the moveable guide vane is limited under the wind attack angles of 0° and -3°.
- The maximum vertical amplitude of the moveable guide vane is much larger than that of the main girder of the bridge in the lock-in region, while the leeward guide vane could play an important role in the VIV suppression when the total stiffness of the guide vane is larger than 19 g/mm.

Although the moveable guide vane with a relatively short length (*i.e.,* 10 mm) and low spring stiffness (*i.e.,* 19 g/mm) could enhance VIV performance of the twin-box girder bridge, the relatively large vertical vibration of the moveable guide vanes may cause instability issues, and thereby further research is required.

CRediT authorship contribution statement

Rui Zhou: Conceptualization, Methodology, Writing – original draft. **Peng Lu:** Software, Formal analysis. **Xiaodong Gao:** Investigation, Data curation. **Yaojun Ge:** Writing – review & editing, Supervision. **Yongxin Yang:** Project administration, Funding acquisition, Visualization. **Lihai Zhang:** Resources, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors gratefully acknowledge the support for the research work jointly provided by the National Natural Science Foundation of China (No. 52178503, 52278311, U2005216, and 51908374), the Shenzhen Science and Technology Program under grant (Nos. JCYJ20220531101609020 and ZDSYS20201020162400001), the Guangdong Natural Science Foundation (No. 2023A1515030148 and 2022A1515010665) and Key Laboratory for Resilient Infrastructures of Coastal Cities (Shenzhen University), Ministry of Education.

References

[1] Ge YJ, Zhao L, Cao JX. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. J Wind Eng Ind Aerod 2022;220:104866.
[2] Zhu J, Xiong ZL, Xiang HY, et al. Ride comfort evaluation of stochastic traffic flow crossing long-span suspension bridge experiencing vortex-induced vibration. J Wind Eng Ind Aerod 2021;219:104794.
[3] Yang YB, Shi K, Mo XQ, et al. Internal instability of thin-walled beams under harmonic moving loads. Thin wall struct 2022;174:109123.
[4] Zhou R, Ge YJ, Yang YX, et al. Nonlinear behaviors of the flutter occurrences for a twin-box girder bridge with passive countermeasures. J Sound Vib 2019;447:221235.
[5] Gao DL, Deng Z, Yang WH, et al. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: A vortex-dynamics approach. J Fluids Struct 2021;105:103348.
[6] Zhou R, Ge YJ, Liu QK, et al. Experimental and numerical studies of wind-resistance performance of twin-box girder bridges with various grid plates. Thin wall struct 2021;166:108088.
[7] Bai H, NaichuanJia NC, Xu GJ, et al. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: Sealed traffic barrier. J Wind Eng Ind Aerod 2020;206:104302.
[8] Larsen A, Savage M, Lafreniere A, et al. Investigation of vortex response of a twin box bridge section at high and low Reynolds numbers. J Wind Eng Ind Aerod 2008;96:934944.
[9] Andersen MS. Operational fluid modal analysis of full-scale pressure and wake flow measurements on the Gjemnessund Bridge. Eng Struct 2021;249:113295.
[10] Kwok KCS, Qin XR, et al. Wind-induced pressures around a sectional twin-deck bridge model: Effects of gap-width on the aerodynamic forces and vortex shedding mechanisms. J Wind Eng Ind Aerod 2012;110:5061.
[11] Wu T, Kareem A. An overview of vortex-induced vibration (VIV) of bridge decks. Front Struct Civ Eng 2012;6:335347.
[12] Zhou R, Ge YJ, Yang YX, et al. A nonlinear numerical scheme to simulate multiple wind effects on twin-box girder suspension bridges. Eng Struct 2019;180:1072–90.
[13] Liu SY, Zhao L, Fang GS, et al. Investigation on aerodynamic force nonlinear evolution for a central-slotted box girder under torsional vortex-induced vibration. J Fluids Struct 2021;106:103380.
[14] Kwon SD, Jung MS, Chang SP. A new passive aerodynamic control method for bridge flutter. J Wind Eng Ind Aerod 2000;86(2–3):187–202.
[15] Bakis K, Massaro M, Williams M, et al. Passive control of bridge wind-induced instabilities by tuned mass dampers and movable flaps. J En Mech 2017;143(9):04017078.
[16] Chen WL, Yang WH, Li H. Self-issuing jets for suppression of vortex-induced vibration of a single box girder. J Fluids Struct 2019;86:213–35.
[17] Larsen A, Esdahl S, Andersen JE, et al. Storebelt suspension bridge vortex shedding excitation and mitigation by guide vanes. J Wind Eng Ind Aerod 2000;88:283296.
[18] Li M, Sun YG, Jing HM, et al. Vortex-Induced Vibration Optimization of a Wide Streamline Box Girder by Wind Tunnel Test. KSCE J Civil Eng 2018;22:51435153.
[19] Hu CX, Zhao L, Ge YJ. Mechanism of suppression of vortex-induced vibrations of a streamlined closed-box girder using additional small-scale components. J Wind Eng Ind Aerod 2019;189:314331.
[20] Ma CM, Li ZG, Meng F, et al. Wind-induced vibrations and suppression measures of the Hong Kong-Zhuhai-Macao Bridge. Wind Struct 2021;32(3):179191.
[21] Li LY, Ge YJ. Experiments of vortex control for central-slotting on long-span bridges. J Huazhong U Sci-Tech 2008;36(12):112115 (in Chinese).
[22] Chen WL, Li H, Hu H. An Experimental Study on the Unsteady Vortices and Turbulent Flow Structures around Twin-Box-Girder Bridge Deck Models with Different Gap Ratios. J Wind Eng Ind Aerod 2014;132:27–36.
[23] Ma CM, Wang JX, Li QS, et al. Vortex-Induced Vibration Performance and Suppression Mechanism for a Long Suspension Bridge with Wide Twin-Box Girder. J Struct Eng 2018;144(11):04018202.
[24] Wu BC, Laima S, et al. Experimental study on characteristics of vortex-induced vibration of a twin-box girder and damping effects. J Fluids Struct 2021;103:103282.
[25] Ge BX, Ma RJ, Li FK, et al. Probabilistic vortex-induced vibration occurrence prediction of the twin-box girder for long-span cable-stayed bridges based on wind tunnel tests. Eng Struct 2022;262:114325.
[26] Yang YX, Zhou R, Ge YJ, et al. Experimental studies on VIV performance and countermeasures for twin-box girder bridges with various slot width ratios. J Fluids Struct 2016;66:476489.
[27] Laima SJ, Li H, Chen WL, et al. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. J Fluids Struct 2018;77:115133.
[28] Wang CQ, Hua XG, Feng ZQ, et al. Experimental investigation on vortex-induced vibrations of a triple-box girder with web modification. J Wind Eng Ind Aerod 2021;218:104783.
[29] Ghosh A, Basu B. A closed-form optimal tuning criterion for TMD in damped structures. Struct control hlth 2007;14(4):681692.

[30] Miguela LFF, Lopez RH, Torii AJ, et al. Robust design optimization of TMDs in vehicle–bridge coupled vibration problems. Eng Struct 2016;126:703–11.
[31] Dai J, Xu ZD, Gai PP. Tuned mass-damper-inerter control of wind-induced vibration of flexible structures based on inerter location. Eng Struct 2019;199:109585.
[32] Tao T, Tam CM. System reliability optimization model for construction projects via system reliability theory. Autom Constr 2012;22:340–7.